

# Analysis of MRI $T_2$ Relaxometry in Extremely Preterm-Born Adolescents

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## Abstract

This report analyses multi-echo  $T_2$  relaxometry data from 134 EPICURE adolescents (81 extremely preterm, 53 full-term), with the aim of testing whether quantitative  $T_2$  and myelin water fraction (MWF) maps can detect persistent white-matter differences associated with extremely preterm birth. On this 10-echo acquisition, the two-compartment fixed- $T_2$  model at 20 and 80 ms is the most informative representation the data support: it is preferred over the mono-exponential model in 95% of white-matter subjects by AICc, while richer three-compartment extensions are not identifiable. Whole-brain group differences are modest, but regional analysis reveals a clear effect in the splenium of the corpus callosum, with elevated  $T_2$  (+4.5 ms) and reduced MWF ( $-0.063$ ) in the preterm group (both  $p < 10^{-4}$ ). In practical terms, the residual myelin differences from extremely preterm birth, nineteen years on, are regionally concentrated rather than spread across the whole brain.

## 1. Introduction

Magnetic resonance image contrast arises from tissue-specific spin relaxation, among which the transverse relaxation time  $T_2$  is sensitive to local water environment and macromolecular content. Quantitative  $T_2$  relaxometry moves beyond qualitative contrast toward tissue parameter maps with biological meaning. When the multi-echo decay is measured with a sufficiently long and densely sampled echo train, a  $T_2$  distribution rather than a single relaxation time can be recovered, and a short component at approximately 10–40 ms has been shown to correspond to water trapped between the myelin bilayers [5, 9]. The fractional area of this component, the myelin water fraction (MWF), correlates quantitatively with Luxol fast blue staining in multiple sclerosis [3] and with diffusion anisotropy in developing and adult brains [6].

In extremely preterm-born cohorts, relaxometry is a promising window on atypical white-matter maturation. The EPICURE cohort, followed into adolescence, shows MWF and  $T_2$  differences relative to term-born peers that are subtle at the whole-brain level but become clearer in specific white-matter tracts [2, 4], consistent with the heterogeneous, region-selective trajectories reported more broadly in preterm-born brain development [8].

Estimating such parameters faithfully is non-trivial. The acquired signal may depart from mono-exponentiality because of multiple water compartments, partial volume, noise floor bias, or motion, and the parameters of multi-compartment models become underdetermined as soon as the number of unknowns approaches the number of echoes [7]. The aim of this report is therefore to work through the relaxometry pipeline on 10-echo EPICURE data, from acquisition quality assessment, through one- and two-compartment fitting, to model comparison, regularised multi-compartment inversion, segmentation-informed priors, and a regional inter-subject analysis. Each step is chosen to trade flex-

ibility against identifiability, and the resulting parameter maps are used to test for group differences between extremely preterm and term-born adolescents.

## 2. Methods

### 2.1 Data and cohort

The adolescent EPICURE dataset comprises multi-echo spin-echo volumes acquired at a shared echo-time schedule of  $TE \in \{13, 16, 20, 25, 30, 40, 50, 85, 100, 150\}$  ms ( $K = 10$  echoes; matrix  $96 \times 96 \times 51$ ). Each subject has a brain mask, a soft tissue segmentation over six classes (unassigned, CSF, cortical GM, WM, deep GM, brainstem) and two anatomical parcellations. Cohort demographics, including gestational age, sex and 19-year neuropsychology scores, are provided in the accompanying `Epicure_psychological.xlsx` file. Subjects lacking a gestational-age or group flag were excluded by inner-join, yielding  $n = 141$  candidates. Four subjects whose TE list did not match the number of image volumes (IDs 44631, 74067, 78004, 98865) were removed. A further three subjects (17065, 48996, 96179) were excluded after visual quality assurance triggered by the monotonicity and log-linearity criteria of Task 1 (§3.1). This yields the frozen analysis cohort used throughout the report:  $n = 134$  (81 extremely preterm, EP; 53 full-term, FT). Six adult-control cases with 19-22-echo schedules serve only as diagnostic examples, because their richer echo trains resolve multi-exponentiality more directly.

### 2.2 Signal-level quality assessment (Task 1)

For every subject, three complementary checks are computed on brain-mask voxels. First, a monotonicity fraction, defined as the proportion of voxels where  $S(TE_{k+1}) \leq S(TE_k)$  holds at every echo pair, identifies acquisitions where noise or motion dominates. Second, a mono-exponentiality score, defined as the

mean absolute log-residual of a log-linear fit averaged across CSF, GM and WM tissue-mean curves, quantifies curvature in log space: a vanishing score is consistent with a single exponential, whereas a positive score indicates multi-exponential behaviour or a noise floor. Third, background-voxel statistics at the last echo are used to estimate the noise level under a Rician model,  $\sigma \approx \bar{S}_{\text{bg}}/\sqrt{\pi/2}$ , and an  $\text{SNR}_k = \bar{S}_k/\sigma$  curve is reported per subject. Partial-volume exposure per tissue class is estimated from the complement of the morphological erosion of the hard segmentation at a single-voxel radius, matching the in-plane voxel size of  $2.5 \times 2.5$  mm reported for the EPICURE protocol. The tissue-mean decay curves used throughout the report are probability-weighted averages over the brain mask, which partially mitigates partial-volume bias at tissue boundaries.

### 2.3 One- and two-compartment models (Tasks 2 and 3)

The one-compartment (mono-exponential) model

$$S(\text{TE}) = S_0 \exp(-\text{TE}/T_2) \quad (1)$$

is fitted with four estimators: (i) a two-point analytic solution from any echo pair, (ii) log-linear ordinary least squares (OLS) on all echoes, (iii) weighted log-linear least squares (WLS) with weights  $w_k = S_k^2$  to approximately compensate for the variance inflation induced by the log transform, and (iv) non-linear least squares (NLLS) with a trust-region reflective solver warm-started from the log-linear estimate. The NLLS fit is bounded by  $S_0 \geq 0$  and  $5 \leq T_2 \leq 3000$  ms to prevent degenerate solutions.

The two-compartment model is

$$S(\text{TE}) = S_0 [v e^{-\text{TE}/T_{2,s}} + (1-v) e^{-\text{TE}/T_{2,l}}], \quad (2)$$

with  $v \in [0, 1]$ . In the free- $T_2$  variant all four parameters are fitted jointly with non-overlapping bounds ( $T_{2,s} \in [5, 50]$  ms,  $T_{2,l} \in [50, 3000]$  ms) to enforce ordering. In the fixed- $T_2$  variant only  $(S_0, v)$  are fitted, with compartment relaxation times held at  $T_{2,s} = 20$  ms and  $T_{2,l} = 80$  ms consistent with values used in literature [7, 2]. The 20 ms component corresponds to myelin water and  $v$  is the MWF. Uncertainty in  $v$  is quantified using a parametric bootstrap with 200 resamples per subject, under a residual-derived Gaussian noise assumption, from which 95% confidence intervals are derived.

### 2.4 Model selection and parameter precision (Task 4)

One- and two-compartment fits on each tissue-mean curve are compared using the small-sample-corrected Akaike information criterion [1],

$$\text{AICc} = 2N + K \log(\text{SSR}/K) + \frac{2N(N+1)}{K-N-1}, \quad (3)$$

with  $K = 10$  echoes and  $N$  the number of estimated quantities including the noise variance ( $N = 3$  for both models:  $(S_0, T_2, \sigma)$  for the one-compartment model, and  $(S_0, v, \sigma)$  for the fixed- $T_2$  bi-exponential variant; AICc difference is driven purely by fit quality). Cohort-level Akaike weights are computed from the mean AICc per

tissue. As an independent check on the bootstrap, a Metropolis-Hastings MCMC chain with a Beta(2, 10) prior on  $v$  and analytic profiling of  $S_0$  is run for 50,000 iterations (10,000 burn-in, thinning factor 20), and the resulting 95% percentile interval is compared against the bootstrap interval on the same voxel.

### 2.5 Multi-compartment models (Task 5)

Two extensions are considered. First, a non-negative least squares (NNLS) inversion over a 101-point log-spaced grid from 10 to 2000 ms recovers a continuous  $T_2$  distribution. Because the linear system is underdetermined ( $J = 101$  basis functions for  $K = 10$  echoes), the plain NNLS solution is sparse and noise-sensitive; a Tikhonov penalty  $\mu^2 \|a\|^2$  is added and  $\mu$  selected by the Whittall  $\chi^2$  criterion [9], namely the smallest  $\mu$  whose regularised misfit does not exceed 1.02 times the plain misfit. The MWF is read as the area of the  $T_2$  spectrum in the 10–40 ms band [6]. Second, a three-compartment fixed- $T_2$  model with  $T_2 \in \{20, 80, 2000\}$  ms, matching Dingwall et al. [2], is fitted with a reparametrisation  $v_1, v_2 \in [0, 1]$  and  $v_3 = 1 - v_1 - v_2$ . A free six-parameter variant is fitted to demonstrate identifiability failure.

### 2.6 Segmentation-informed priors (Task 6)

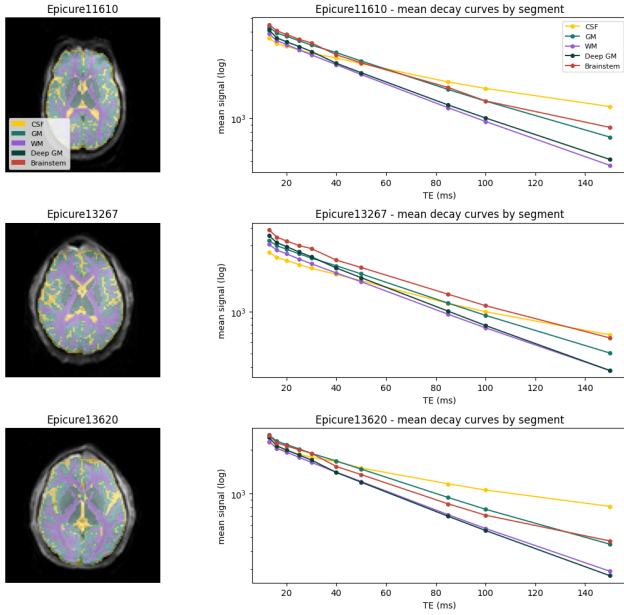
Data-derived cohort means  $\mu_t$  and standard deviations  $\sigma_t$  per tissue  $t \in \{\text{CSF}, \text{GM}, \text{WM}\}$  are obtained from the Task 3 tissue-mean fits. At each voxel, the soft-segmentation probabilities  $p_t$  are used to construct a voxel-specific Gaussian prior on  $v$ ,

$$\mu_v = \sum_t p_t \mu_t, \quad \sigma_v^2 = \sum_t p_t \sigma_t^2, \quad (4)$$

where deep GM and brainstem probabilities are folded into the GM class and a floor  $\sigma_v \geq 0.02$  is imposed to prevent near-singular priors in near-pure CSF voxels. Three voxelwise strategies are then compared on the same slice: plain NLLS with a fixed initialisation  $v_{\text{init}} = 0.10$ ; a warm-start variant with  $v_{\text{init}} = \mu_v$ ; and a maximum-*a-posteriori* (MAP) variant that augments the NLLS residuals with a Gaussian penalty  $(v - \mu_v)/\sigma_v$  scaled against a subject-wise noise estimate  $\sigma_{\text{noise}}$  derived from the WM tissue-mean residual. Independently of the prior, spatial smoothness within the WM mask is tested by Gaussian smoothing with a 2-voxel FWHM applied only inside the tissue, preventing leakage across boundaries.

### 2.7 Inter-subject and regional analysis (Task 7)

Two parameters are derived per subject: a whole-WM  $T_2$  from mono-exponential NLLS and a whole-WM MWF from the fixed- $T_2$  bi-exponential fit. The same two fits are then applied at regional level using the supplied parcellations. The WM parcellation (`par2.nii.gz`) defines the posterior and anterior limbs of the internal capsule (PLIC, ALIC) and the genu and splenium of the corpus callosum. For each region, a tissue-mean decay curve is constructed by weighting voxel signals with the WM tissue probability. The cortical GM parcellation (`par1.nii.gz`) defines twelve



**Figure 1:** Central slices with tissue-segmentation overlays and corresponding segment-wise mean decay curves for three representative EPICURE subjects.

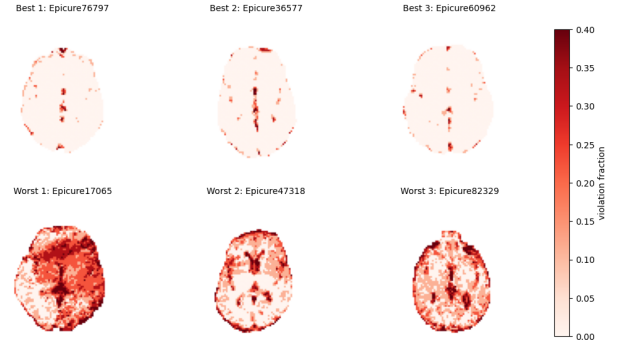
regions, bilateral frontal, temporal, parietal, occipital, cingulate, and insular cortex, and the corresponding regional decay curves are weighted by cortical GM tissue probability. Group comparisons (EP versus FT) use Welch’s independent-samples  $t$ -test, Cohen’s  $d$  for effect size, and a 2000-resample percentile bootstrap for the 95% confidence interval on the mean difference. Bonferroni correction is applied separately within the WM and GM analyses, at a family-wise  $\alpha = 0.05$  level. For WM, this gives 8 comparisons (2 metrics  $\times$  4 regions;  $\alpha = 0.0063$ ). For GM, it gives 24 comparisons (2 metrics  $\times$  12 regions;  $\alpha = 0.0021$ ). Cognitive outcome is examined via Spearman rank correlation between whole-WM parameters and 19-year Full-Scale IQ, within and across groups.

### 3. Results

#### 3.1 Signal quality and acquisition integrity

Tissue-mean decay curves follow the expected semilog ordering: CSF decays slowest, WM and deep GM fastest, with cortical GM and brainstem in between (Figure 1). Semilog curves for WM and GM are visibly concave rather than strictly linear, which is the first qualitative indication that a single  $T_2$  does not fully capture the observed decay and the mono-exponential model is only an approximation of the underlying signal decay.

The monotonicity check returns a cohort distribution of strictly monotonic voxels from 12% (worst, Epicure17065) to 89% (best, Epicure76797), with a mean per-echo violation rate below 5% in the bulk of the cohort. Inspection of the spatial maps shows a clear contrast between expected boundary-dominated patterns and more diffuse, acquisition-related violation patterns (Figure 2). In the best subjects, violations are sparse and mainly boundary-concentrated, which is the expected signature of partial-volume mixing and



**Figure 2:** Voxel-wise monotonicity-violation fraction on the central axial slice for the three best and three worst subjects in the cohort. The maps show, for each voxel, the fraction of consecutive echo pairs for which the signal increases rather than decreases

**Table 1:** Cohort  $T_2$  summaries per tissue across the preterm analysis cohort ( $n = 134$ ), reported as median (range) in ms.

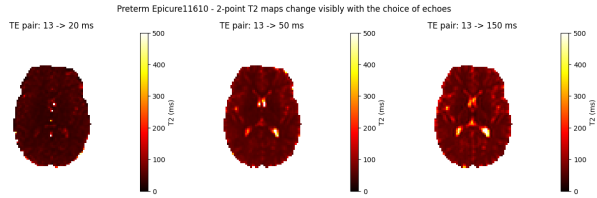
| Tissue | OLS                | WLS                | NLLS               |
|--------|--------------------|--------------------|--------------------|
| CSF    | 116.5 (91.7–316.9) | 107.4 (84.1–299.7) | 106.6 (83.4–301.6) |
| GM     | 78.0 (73.4–85.6)   | 75.0 (70.8–81.7)   | 75.0 (70.4–81.6)   |
| WM     | 67.5 (64.2–85.9)   | 64.5 (61.2–80.6)   | 64.4 (61.1–80.3)   |

noise-floor effects. In the worst subjects, however, violations are substantially more diffuse and extend well into the brain interior, indicating that the poorest scans are unlikely to be explained by boundary effects alone and instead reflect genuine acquisition degradation. The mono-exponentiality score has a narrow distribution (mean  $0.025 \pm 0.004$ , range 0.019–0.049), with the worst subject again Epicure17065. The Rician noise estimate is small relative to tissue signal: median  $\sigma = 22.0$ , median SNR at the first echo 161.6, median SNR at the last echo 29.9, with no subject dropping below an SNR of 3 at any echo. This indicates that, at the group level, the Rician noise floor bias on late echoes can be neglected [7]. Partial-volume exposure, estimated as the fraction of each tissue that is eroded away at a 1-voxel radius, is substantial (median CSF 96.4%, GM 87.1%, WM 70.8%); this motivates the use of probability-weighted tissue-mean curves, rather than hard-segmentation means, throughout the remainder of the report. After flagging the worst decile on either criterion (25/137 subjects) and visually inspecting each, three subjects (17065, 48996, 96179) are excluded on combined evidence of acquisition artefacts, leaving the frozen  $n = 134$  analysis cohort.

#### 3.2 One-compartment fitting

The two-point analytic estimator is strongly dependent on the chosen echo pair. On the WM mean curve of Epicure11610, increasing the second echo from 16 to 150 ms changes the recovered  $T_2$  from 28.0 to 64.9 ms, whereas the all-echo log-linear estimate is 65.7 ms (Figure 3). This sensitivity arises because each two-point estimate is determined by only the line through the two sampled echoes, and short echo pairs weight the fast-decaying component disproportionately. Using all echoes reduces this dependence on echo selection and yields physically sensible, tissue-ordered  $T_2$  values across the preterm cohort (Table 1).

Across the cohort, WLS and NLLS agree to within



**Figure 3:** Two-point  $T_2$  maps for Epicure11610 obtained from three different echo pairs.

0.5 ms in GM and WM (Table 1), whereas OLS consistently yields larger  $T_2$  estimates, most markedly in CSF where the range extends to 316.9 ms. This reflects the well-known bias of unweighted log-linear fitting: the logarithm magnifies the residual weight of low-signal, late-TE measurements, where CSF retains signal and noise variance becomes dominant. NLLS agrees closely with WLS, particularly in GM and WM, indicating that the limitations of the one-compartment model are not specific to a single fitting algorithm. Computationally, NLLS is substantially slower than log-linear OLS. At the whole-brain level the projected runtime is 102.4 s versus 1.91 s (approximately 54x longer).

The most informative output of the one-compartment analysis is the residual structure. For NLLS fits on WM and GM tissue-mean curves, residuals are systematically positive at short TE, negative at intermediate TE, and positive again at long TE (Figures 4 and 5), a pattern reproducible across subjects and groups. This is the classical fingerprint of a missing short- $T_2$  component and motivates the bi-exponential analysis.

### 3.3 Two-compartment fitting and parameter precision

On the 22-echo adult diagnostic case, the free- $T_2$  bi-exponential fit recovers  $(S_0, v, T_{2,s}, T_{2,l}) = (1558.2, 0.119, 14.0 \text{ ms}, 69.6 \text{ ms})$  and is numerically invariant across the initialisations  $v_{\text{init}} \in \{0.05, 0.10, 0.20, 0.40\}$ . This behaviour indicates that, with a sufficiently rich echo train, the four-parameter model is identifiable on a tissue-mean curve. On the 10-echo Epicure11610 WM curve, however, the same free- $T_2$  fit collapses to  $T_{2,s} = 5 \text{ ms}$ , the lower bound, with  $v = 0.489$  regardless of initialisation, which is not an informative solution. The fixed- $T_2$  bi-exponential model with  $(T_{2,s}, T_{2,l}) = (20, 80) \text{ ms}$  remains stable on the same curve and yields  $v = 0.330$ , supporting the use of fixed compartment times on the adolescent 10-echo acquisition.

A further sanity check compares the fit on the tissue-mean curve with the median of voxelwise fits for the same subject. In WM, the agreement is close (0.330 versus 0.348, 17,480/17,480 voxels converged), while in GM the offset is somewhat larger but remains qualitatively consistent (0.127 versus 0.159, 28,894/28,920 voxels converged). This supports the use of tissue-mean curves as cohort summaries, particularly for WM, while also indicating that voxelwise heterogeneity is not entirely removed by the mean-curve approximation.

Applied to every (subject, tissue) combination in the adolescent analysis cohort, the fixed- $T_2$  model yields

**Table 2:** Cohort fixed- $T_2$  MWF statistics in the adolescent analysis cohort ( $n = 134$ ) with  $T_{2,s} = 20 \text{ ms}$  and  $T_{2,l} = 80 \text{ ms}$ . Median CI width is from 200 parametric bootstrap resamples per fit.

| Tissue | Mean $v$ | SD $v$ | Median 95% CI width |
|--------|----------|--------|---------------------|
| CSF    | 0.000    | 0.003  | 0.280               |
| GM     | 0.123    | 0.037  | 0.116               |
| WM     | 0.282    | 0.039  | 0.100               |

cohort mean MWF values of  $0.282 \pm 0.039$  in WM,  $0.123 \pm 0.037$  in GM, and  $0.000 \pm 0.003$  in CSF, pooled across groups (Table 2). The ordering is biologically plausible, with the largest short- $T_2$  fraction in WM, a smaller value in GM, and an estimate that is essentially zero across subjects in CSF. The CSF behaviour is expected, because a model with a fixed 20 ms short component is not an appropriate description of a predominantly long- $T_2$  fluid compartment.

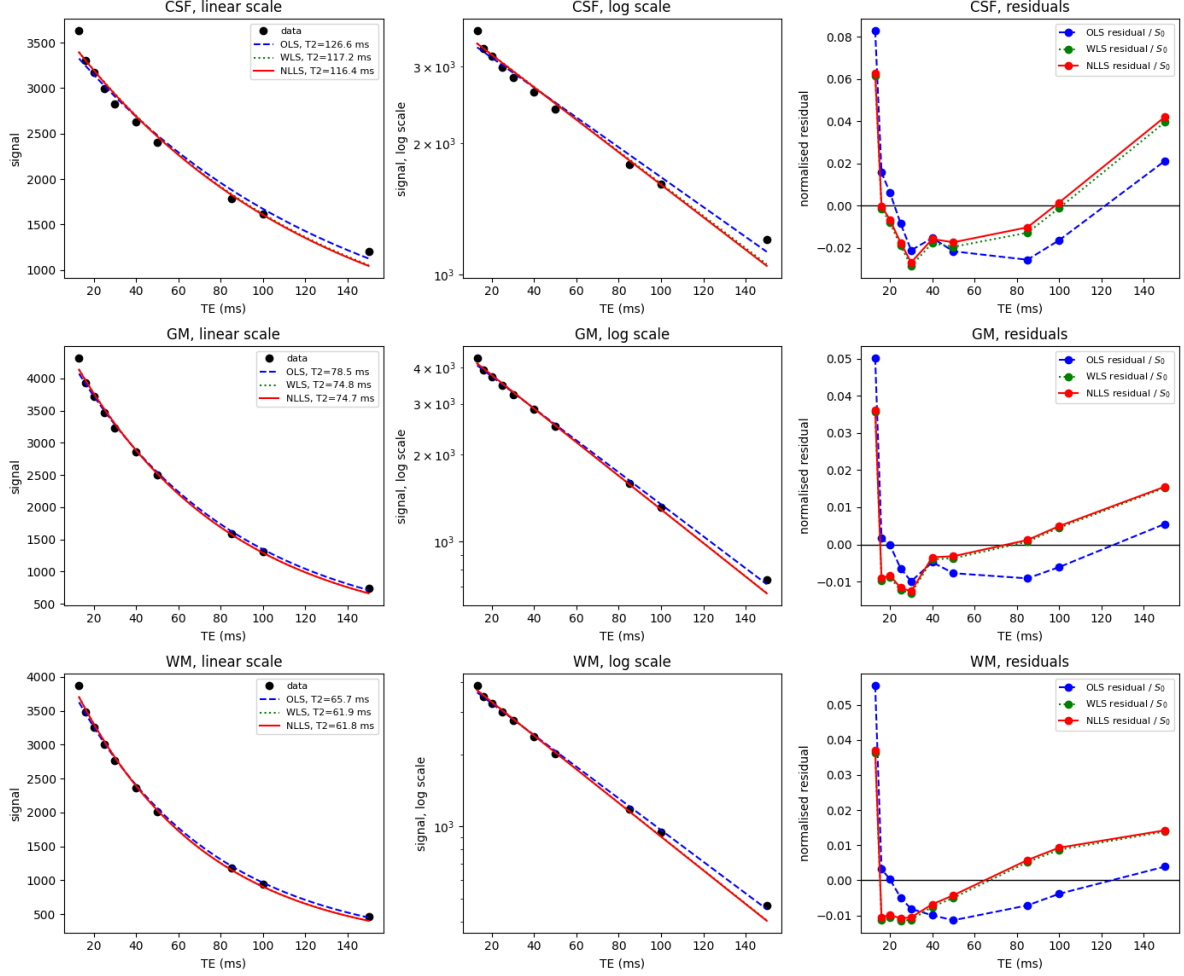
To quantify uncertainty, a parametric bootstrap with 200 resamples was run for each fit. The resulting median 95% interval widths are 0.100 in WM, 0.116 in GM, and 0.280 in CSF. The CSF intervals are wide because the fit sits at or near the boundary  $v = 0$  and the likelihood is correspondingly flat in that direction, which is the expected signature of model mis-specification. By contrast, WM and GM estimates are much better constrained, although the single-subject interval widths remain non-negligible and should not be over-interpreted as high-precision voxel-level myelin measurements.

### 3.4 Model selection and uncertainty quantification

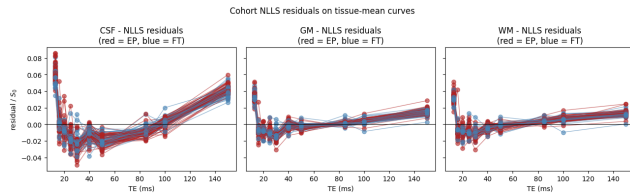
Model comparison was performed between the one-compartment NLLS fit and the fixed- $T_2$  two-compartment fit using the small-sample-corrected Akaike information criterion, AICc. In the notebook implementation, both models are counted with the same effective number of estimated quantities ( $k = 3$ , including the residual variance), because the short and long relaxation times of the bi-exponential model are fixed at 20 and 80 ms. Accordingly, the AICc difference here reflects fit quality alone: negative  $\Delta\text{AICc} = \text{AICc}_{2c} - \text{AICc}_{1c}$  favours the two-compartment model, whereas positive values favour the one-compartment model [1].

The cohort-level pattern is clear (Table 3, Figure 8). In WM, the two-compartment model is preferred in 95% of subjects, with median  $\Delta\text{AICc} = -4.29$  and cohort Akaike weight  $w_{2c} = 0.90$ . In GM, the evidence is similarly strong: 98% of subjects favour the two-compartment model, with median  $\Delta\text{AICc} = -4.95$  and  $w_{2c} = 0.92$ . In CSF, by contrast, the one-compartment model is preferred in all subjects (median  $\Delta\text{AICc} = +14.94$ ,  $w_{1c} = 1.00$ ). This is the expected outcome, because CSF is dominated by a single long- $T_2$  component and a fixed 20 ms short component only worsens the fit.

Parameter uncertainty on the WM myelin-water fraction  $v$  was then examined on the Epicure11610 WM tissue-mean curve using two complementary methods: a parametric bootstrap with 1000 resamples and a Metropolis–Hastings MCMC sampler with a Beta(2, 10)



**Figure 4:** Representative tissue-mean one-compartment fits for Epicure11610. Rows show CSF, GM and WM; columns show linear-scale fits, log-scale fits, and normalised residuals for OLS, WLS and NLLS.

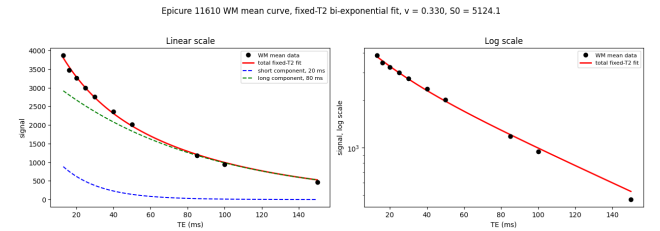


**Figure 5:** Per-echo normalised residuals of the NLLS one-compartment fit on tissue-mean curves across the cohort, shown separately for CSF, GM and WM (EP in red, FT in blue).

**Table 3:** AICc-based model comparison between the one-compartment and fixed- $T_2$  two-compartment models across the adolescent analysis cohort ( $n = 134$ ). Negative  $\Delta\text{AICc}$  favours the two-compartment model.

| Tissue | Median $\Delta\text{AICc}$ | 2-comp preferred | $w_{1c}$ | $w_{2c}$ |
|--------|----------------------------|------------------|----------|----------|
| WM     | -4.29                      | 95%              | 0.10     | 0.90     |
| GM     | -4.95                      | 98%              | 0.08     | 0.92     |
| CSF    | +14.94                     | 0%               | 1.00     | 0.00     |

prior on  $v$ . The two procedures give closely matching results. The bootstrap returns  $v = 0.330$  with 95% interval  $[0.282, 0.377]$ , while the MCMC posterior has median  $v = 0.323$  with 95% credible interval  $[0.272, 0.369]$ . The posterior is shifted slightly toward lower values, which is expected given the asymmetric prior, but the difference is modest relative to the overall interval width.

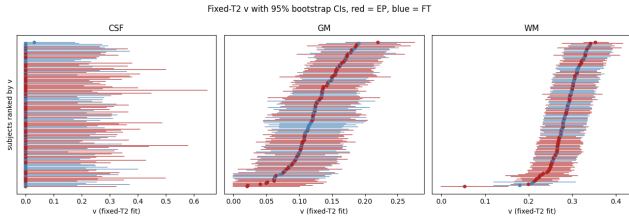


**Figure 6:** Representative fixed- $T_2$  bi-exponential fit on the Epicure11610 WM mean curve. Left, linear-scale fit with decomposition into the 20 ms and 80 ms components. Right, the same fit on a log scale.

### 3.5 Multi-compartment analyses

A continuous  $T_2$  distribution was estimated with regularised NNLS. This regularisation is necessary because the inversion is strongly underdetermined: the spectrum is represented on a dense 101-point  $T_2$  grid, whereas the adolescent dataset contains only 10 echoes. Without regularisation, NNLS tends to concentrate the signal into a few unstable spikes. The Tikhonov penalty therefore stabilises the inversion by spreading the signal over neighbouring  $T_2$  bins, while the Whittall  $\chi^2$  rule keeps the increase in misfit small.

The resulting spectra are most useful as a qualitative description of multi-component signal behaviour rather



**Figure 7:** Subject-level fixed- $T_2$  bi-exponential  $v$  estimates with 95% bootstrap intervals, shown separately for CSF, GM, and WM and coloured by group.

than as a precise count of distinct compartments. In the preterm cohort, GM and WM are both dominated by an intermediate peak around 50–150 ms, while CSF shows a broader distribution shifted towards longer  $T_2$  values and retains visible long- $T_2$  weight (Figure 10). The tissue separation is only moderate, which is expected for tissue-mean curves derived from a 10-echo acquisition and is further blurred by partial-volume mixing.

For comparison, the same regularised NNLS analysis was applied to an adult diagnostic control case with 22 echoes. The richer echo sampling yields cleaner and more sharply defined spectra, but the same overall tissue ordering remains visible: GM and WM are centred on an intermediate component, whereas CSF is broader and shifted towards longer  $T_2$ . Together, these results support a qualitatively plausible multi-compartment interpretation, while also showing that the 10-echo adolescent data do not support fine spectral resolution.

In WM, the MWF estimate depends on the model used. The regularised NNLS spectrum gives a lower WM MWF than the parametric models, with cohort mean  $0.198 \pm 0.040$ , whereas the fixed- $T_2$  two-compartment and three-compartment models give almost identical values,  $0.282 \pm 0.039$  and  $0.283 \pm 0.034$ , respectively. This suggests that the continuous-spectrum NNLS approach gives a more conservative estimate of the short- $T_2$  fraction, while adding a third fixed compartment does not materially change the inferred WM myelin fraction.

Model comparison indicates that the extra compartment is not supported by the 10-echo data. In WM, the cohort-average Akaike weights are  $(w_1, w_2, w_3) = (0.10, 0.86, 0.04)$  for the one-, two-, and three-compartment models, and the per-subject difference  $\Delta\text{AICc}(3-2)$  has median +6.00. Thus, the fixed- $T_2$  two-compartment model is strongly preferred over the three-compartment extension.

This conclusion is reinforced by the free- $T_2$  three-compartment fit. On the Epicure11610 WM mean curve, all tested initialisations converge to the same boundary solution, with  $T_{2,s} = 5$  ms and  $T_{2,l} = 3000$  ms. This does not imply that a third biological water pool is absent, but rather that the current 10-echo acquisition does not contain enough information to estimate a free three-compartment model reliably. Overall, the data support multi-component behaviour, but the fixed- $T_2$  two-compartment model remains the most interpretable summary.

### 3.6 Segmentation-informed priors and spatial smoothing

At voxel level, warm-starting from the segmentation prior produces essentially the same WM distribution as the no-prior baseline. On the Epicure11610 demonstration slice, both approaches give median  $v = 0.373$  and interquartile range  $[0.313, 0.449]$  (Figure 11). Therefore, changing only the initial value of  $v$  does not alter the fitted solution. The MAP prior has a clearer effect. On the same slice, the WM median shifts to 0.345, and the interquartile range narrows to  $[0.298, 0.401]$ . The main benefit is therefore not faster convergence, but a reduction in extreme voxel estimates.

This pattern generalises across the cohort. Relative to the no-prior fit, the MAP strategy reduces noise and boundary-sticking without shifting the cohort median of WM MWF by more than 0.05 ( $\Delta\text{median} = -0.046$ ,  $\Delta\text{SD} = -0.018$ , and  $\Delta\text{boundary}\% = -0.53$  percentage points, averaged over  $n = 134$  subjects). The prior thus produces smoother and more stable WM maps, while shifting the median only modestly at cohort level.

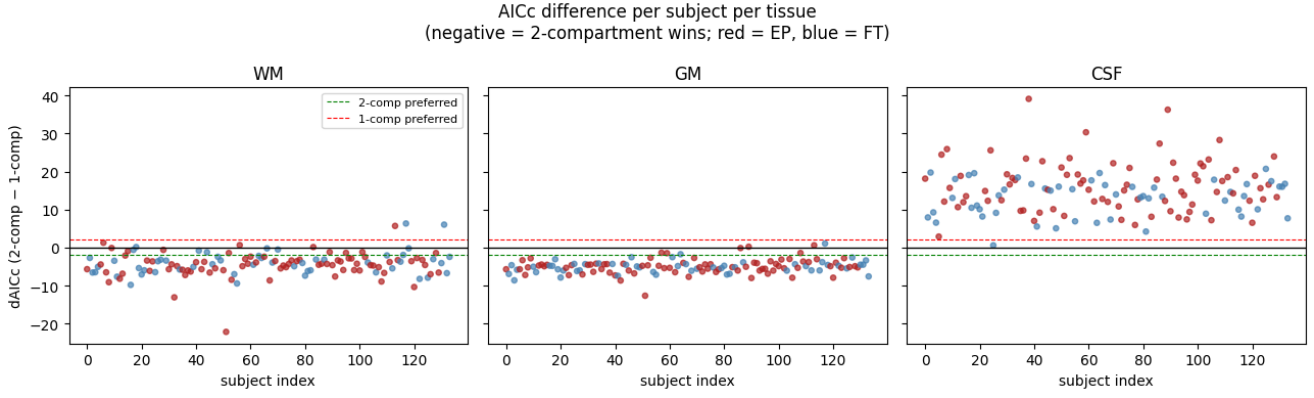
As a separate post-processing step, Gaussian smoothing within the WM mask reduces voxelwise SD from 0.097 to 0.063 on the demonstration slice. This confirms that neighbouring WM voxels contain redundant information that can be used to suppress noise, although the stronger result in this task is the segmentation-informed MAP fit itself.

### 3.7 Inter-subject and regional variation

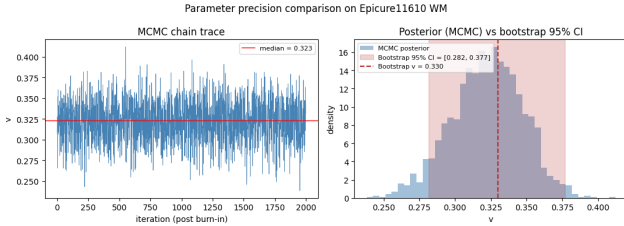
At the whole-WM level, EP subjects show the expected direction of effect: slightly higher  $T_2$  and slightly lower MWF than FT subjects (Table 4, top two rows). The differences are small (Cohen’s  $d$  around 0.4 in both directions) but statistically significant, with mean shifts of +0.91 ms for  $T_2$  and  $-0.014$  for MWF (Figure 12). The effect attenuates under sex stratification (females:  $p = 0.052$ ,  $d = +0.46$ ; males:  $p = 0.17$ ,  $d = +0.32$ ).

The same conclusion emerges from the clinical variables. Full-Scale IQ is strongly separated between groups, yet within the EP group neither whole-WM  $T_2$  nor MWF shows a meaningful association with FSIQ, and neither metric correlates with gestational age. Whole-WM summaries therefore detect a small EP-versus-FT shift, but do not explain the much larger inter-subject variation in cognitive outcome or birth history.

Regional analysis on the internal capsule and corpus callosum labels (Figure 13) sharpens the picture. The splenium of the corpus callosum shows the largest group contrast in both metrics and is the only region where both  $T_2$  (diff +4.50 ms,  $p < 0.0001$ ,  $d = +0.69$ ) and MWF (diff  $-0.063$ ,  $p < 0.0001$ ,  $d = -0.75$ ) survive Bonferroni correction at a family-wise  $\alpha = 0.05$ . The genu of the corpus callosum shows a MWF reduction in EP that is significant at uncorrected  $\alpha = 0.05$  but does not survive Bonferroni ( $p_{\text{raw}} = 0.009$ ,  $p_{\text{bonf}} = 0.074$ ). The PLIC shows an uncorrected MWF elevation in EP that has no Bonferroni survival. Full regional statistics are reported in Table 4.



**Figure 8:** Per-subject  $\Delta\text{AICc}$  for the fixed- $T_2$  two-compartment model relative to the one-compartment model, shown separately for WM, GM, and CSF. Negative values favour the two-compartment model.



**Figure 9:** Parameter precision for the fixed- $T_2$  WM myelin-water fraction on Epicure11610. Left, post-burn-in MCMC trace. Right, MCMC posterior overlaid with the 95% parametric-bootstrap interval and bootstrap point estimate.

The parallel cortical GM analysis across the twelve `par1.nii.gz` regions yielded no Bonferroni survivors within the GM family. Five regions reached uncorrected significance at  $p < 0.05$ . Four showed elevated  $T_2$  in EP, namely bilateral cingulate cortex, right insular cortex, and left occipital cortex ( $p \in [0.021, 0.037]$ ), while the right insular cortex also showed reduced MWF ( $p = 0.018$ ,  $d = -0.41$ ). None of these effects survived the family-wise threshold of  $\alpha = 0.0021$ , supporting the interpretation that the persistent preterm signal at 19 years is concentrated in callosal WM rather than in cortical GM.

This pattern is consistent with the preterm-adolescent literature. The weak whole-brain signal is weak, whereas regional analyses, particularly in the corpus callosum, are more sensitive to persistent myelination differences [4]. The underlying between-subject heterogeneity that these group-averaged designs tend to under-detect is discussed by [8].

## 4. Discussion

### 4.1 Principal findings

This study set out to establish how informative multi-echo  $T_2$  relaxometry can be on a 10-echo acquisition and, against that background, whether quantitative  $T_2$  and MWF maps reveal residual white-matter differences nineteen years after extremely preterm birth. Three principal findings emerge.

First, the two-compartment fixed- $T_2$  model at 20 and 80 ms is the most information-dense representation that the adolescent EPICURE schedule supports. The mono-exponential model is rejected in 95% of WM and 98%

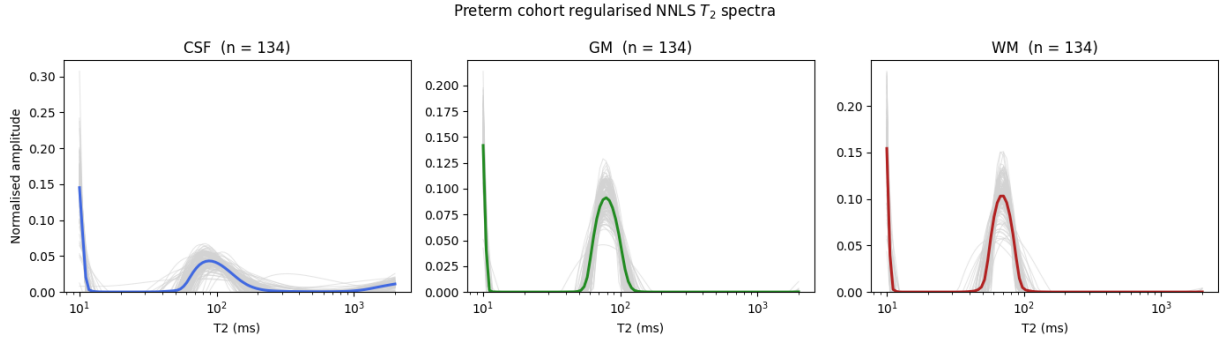
of GM subjects by AICc, and the pattern of residuals, positive at short TE, negative at intermediate TE, and positive again at long TE, is the classical fingerprint of an unmodelled short- $T_2$  component. At the other end, both the free- $T_2$  bi-exponential variant and the three-compartment extensions are not identifiable on 10 echoes: the four-parameter fit collapses to its  $T_{2,s}$  lower bound regardless of initialisation, and the three-compartment Akaike weight in WM is only 0.04. Adding flexibility to the model therefore buys no additional interpretable information on this acquisition.

Second, parameter precision at the single-subject level is acceptable but not spectacular. Parametric bootstrap and MCMC intervals agree to within 0.01 in  $v$  on the reference WM curve, and the data-derived MAP prior narrows the voxelwise WM distribution, interquartile range 0.313–0.449 without prior versus 0.298–0.401 with MAP, without shifting the cohort median by more than 0.05. This is practically important: for a cohort of 134 subjects the bootstrap is tractable and matches the more expensive Bayesian sampler, and the segmentation-informed prior improves map quality without introducing a cohort-level bias.

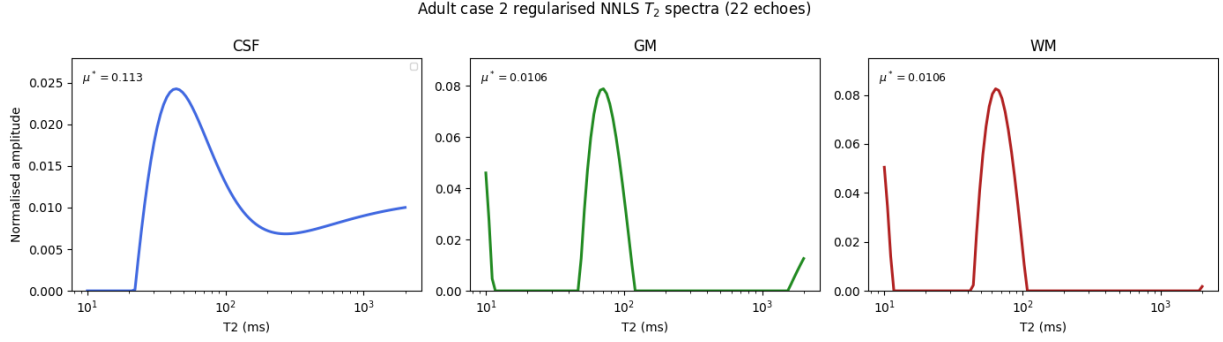
Third, and most importantly from a clinical standpoint, the EP-versus-FT difference is real but regionally concentrated. At the whole-WM level,  $T_2$  is elevated (+0.91 ms) and MWF is reduced (−0.014) in the EP group, with Cohen’s  $d \approx 0.4$ . These effects are in the expected direction and statistically significant, but they do not survive stratification by sex. The regional analysis tells a sharper story: the splenium of the corpus callosum shows +4.5 ms in  $T_2$  and −0.063 in MWF between groups, both  $p < 10^{-4}$  and  $d \approx 0.7$ , surviving Bonferroni correction across eight tests. In practical terms, less myelin in these posterior fibres likely means that communication between the two halves of the brain is a bit less efficient. This may slightly reduce how well posterior brain regions work together, rather than pointing to a general problem across all white matter.

### 4.2 Summary

On this 10-echo acquisition, the fixed- $T_2$  two-compartment model at 20 and 80 ms is the best-supported description of the multi-echo decay, whereas



(a) Regularised NNLS spectra across the preterm cohort ( $n = 134$ ). Grey curves are individual subjects and the coloured line is the tissue mean.

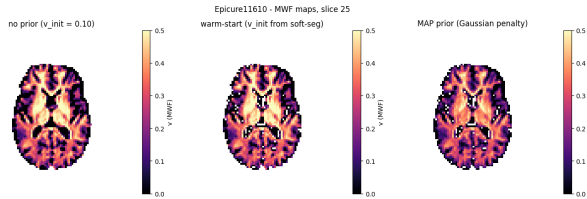


(b) Regularised NNLS spectra for adult control case 2 (22 echoes), shown for CSF, GM and WM tissue-mean curves.

**Figure 10:** Regularised NNLS  $T_2$  spectra for the adolescent cohort and a higher-resolution adult diagnostic case. The cohort data show broader and less clearly separated tissue spectra, whereas the adult control spectra are cleaner because of the richer echo sampling.

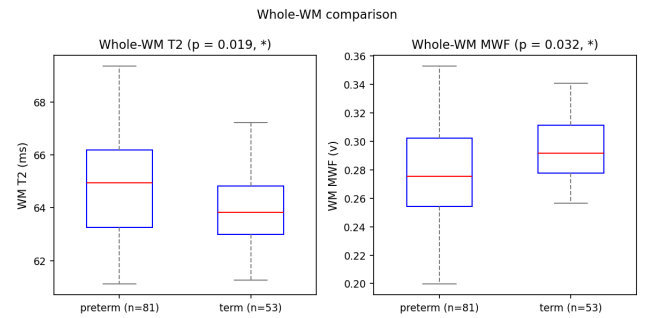
**Table 4:** Group comparisons on whole-WM and regional  $T_2$  and MWF in the preterm analysis cohort ( $n_{EP} = 81$ ,  $n_{FT} = 53$ ). Raw  $p$ -values are from Welch two-sample tests;  $p_{\text{bonf}}$  is Bonferroni-corrected across the eight regional comparisons.

| Metric     | Region      | EP    | FT    | diff   | 95% CI           | $t$   | $p$         | $p_{\text{bonf}}$ |
|------------|-------------|-------|-------|--------|------------------|-------|-------------|-------------------|
| $T_2$ (ms) | whole WM    | 64.96 | 64.05 | +0.91  | [+0.16, +1.67]   | +2.38 | 0.019       | —                 |
| MWF        | whole WM    | 0.277 | 0.290 | −0.014 | [−0.026, −0.001] | −2.17 | 0.032       | —                 |
| $T_2$ (ms) | PLIC        | 59.97 | 62.29 | −2.32  | [−4.33, −0.71]   | −2.57 | 0.012       | 0.095             |
| $T_2$ (ms) | Genu CC     | 61.33 | 58.29 | +3.05  | [−0.51, +6.35]   | +1.71 | 0.090       | 0.719             |
| $T_2$ (ms) | ALIC        | 52.30 | 51.13 | +1.16  | [−0.03, +2.48]   | +1.78 | 0.078       | 0.626             |
| $T_2$ (ms) | Splenium CC | 65.51 | 61.00 | +4.50  | [+2.64, +6.49]   | +4.58 | $< 10^{-4}$ | $< 10^{-4} * *$   |
| MWF        | PLIC        | 0.374 | 0.345 | +0.030 | [+0.007, +0.056] | +2.46 | 0.016       | 0.126             |
| MWF        | Genu CC     | 0.354 | 0.404 | −0.049 | [−0.087, −0.014] | −2.64 | 0.009       | 0.074             |
| MWF        | ALIC        | 0.478 | 0.496 | −0.018 | [−0.039, +0.000] | −1.80 | 0.075       | 0.602             |
| MWF        | Splenium CC | 0.294 | 0.357 | −0.063 | [−0.089, −0.038] | −4.85 | $< 10^{-4}$ | $< 10^{-4} * *$   |



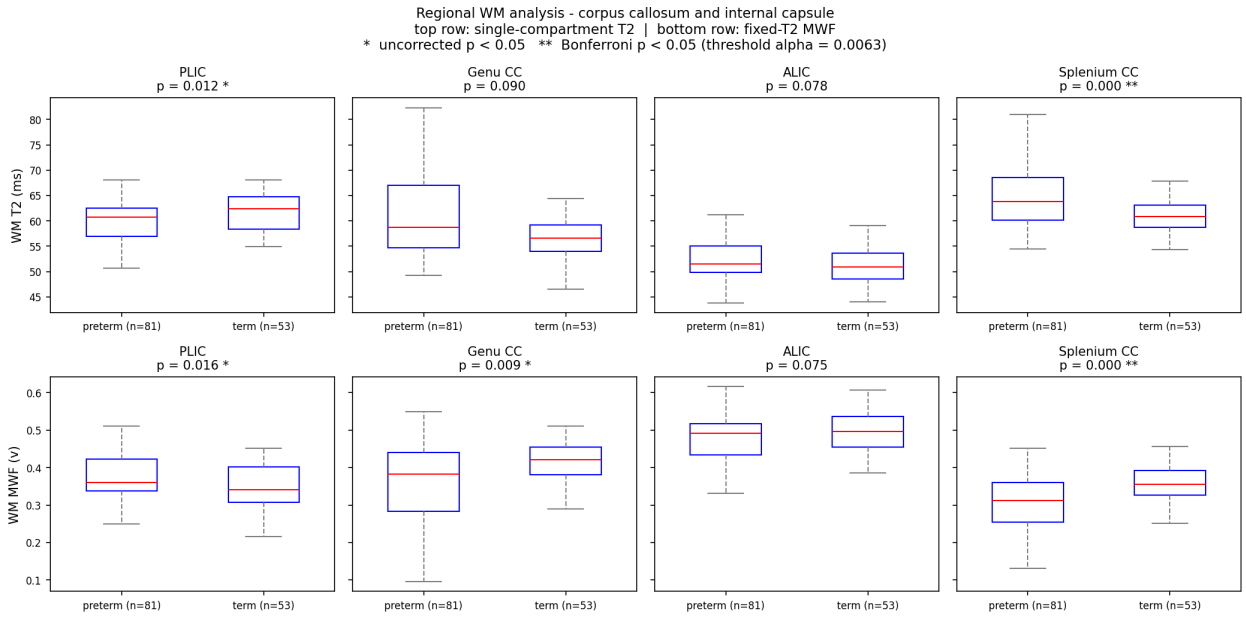
**Figure 11:** Voxelwise WM MWF maps on a central axial slice of Epicure11610 under three fitting strategies: no prior, warm-start only, and MAP prior. The warm-start map is visually indistinguishable from the no-prior result, whereas the MAP prior produces a smoother WM distribution with fewer extreme voxels.

richer models are either unidentifiable or rejected by AICc. Under this model, extremely preterm-born adolescents differ significantly from term-born peers in whole-WM  $T_2$  and MWF ( $p = 0.019$  and  $p = 0.032$ , respectively), but these effects remain modest in magnitude, with Cohen’s  $d$  around 0.4 in both directions and absolute shifts of only +0.91 ms in  $T_2$  and −0.014 in MWF. These whole-WM differences are an order of



**Figure 12:** Whole-WM  $T_2$  and MWF in EP and FT subjects, shown for the full cohort and after stratification by sex. The direction of effect is preserved in each panel, but the separation remains modest.

magnitude smaller than those observed in the splenium of the corpus callosum and do not survive stratification by sex, indicating that the group contrast is detectable at cohort level but not robust to subgroup analysis. The clinically interpretable signal is therefore regionally rather than globally distributed, with the splenium



**Figure 13:** Regional white-matter  $T_2$  and MWF in EP and FT subjects for the PLIC, genu corpus callosum, ALIC, and splenium corpus callosum. The splenium is the only region where both metrics survive Bonferroni correction in the expected direction.

emerging as the clearest locus of persistent alteration, consistent with the stronger callosal effect reported by Laureano et al. [4].

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